Phantom Road Gaps

Reference bookkeeping makes up most of the road-coverage gap, and it bends the scorecard built to reveal inequity

Team Yesternity

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Entered for: Best Bias Discovery (Part A) and Best Documentation (Part B). **Leaderboard submissions this writeup describes:** `EVGrZBHt` (E134, public RMSE 0.05556589, private 0.056132715) and `E3u62GkV` (E136, public RMSE 0.055506023, private 0.05613082), our two selected finals. Both are built only from the challenge package.

Abstract. The Bias Bounty coverage-gap score compares Overture's map of each census tract with hidden reference layers, and a scorecard then compares the score across rural, tribal, vulnerable and hazard-exposed places. We show that the road part of the score mostly measures how the reference is counted, and that this bends the scorecard itself. Two counting rules do it: a highway along a tract boundary is counted in full in both tracts, and a stretch that carries two route numbers or a second name is counted again. Measured from the challenge package alone and tested against the hidden scores, they make up 72% of the mean road gap of the 6,593 road-scored tracts (55–80% across every weighting we tried; 0.23 is added in urban Oklahoma, 0.07 in urban Arizona). Counted once, the rural-to-urban ratio of the score rises from 2.34 to 2.73 and the wildfire-hazard ratio from 1.67 to 1.81, while the social-vulnerability ratio falls from 1.16 to 1.01: the reported disadvantage of rural and fire-exposed places is understated, and nearly all of the reported social-vulnerability disparity is bookkeeping. On a worklist of the 660 worst-scored tracts, 321 are there for highways already on the map, and the 321 they push off it are 71% rural, 58% tribal and 83% above the median wildfire hazard. Public Census and FEMA data agree: those 321 have 1.8 times the share of mobile homes of the tracts that crowd them out, and FEMA rates three times as many high for wildfire risk. Every number is recomputed by a named script; 33 are checked in a second from a few megabytes.

The finding in four numbers

72% of the mean road gap is bookkeeping, not missing roads (55–80% under any weighting; Sec. 3).

2.34 → 2.73: the scorecard's rural ratio counted once; social vulnerability 1.16 → 1.01 (Fig. 1).

321 of 660 worst-scored tracts are on the list for highways that are mapped already (Fig. 5).

71% rural, 58% tribal: the 321 tracts the bookkeeping pushes off that list (Table 6).

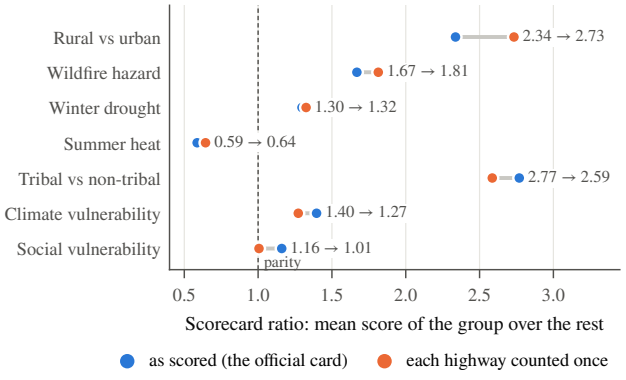


Figure 1: The scorecard on our final, as scored (the card the leaderboard prints) and counted once: rural and wildfire disadvantage is understated; the social-vulnerability gap is mostly bookkeeping (§4.1).

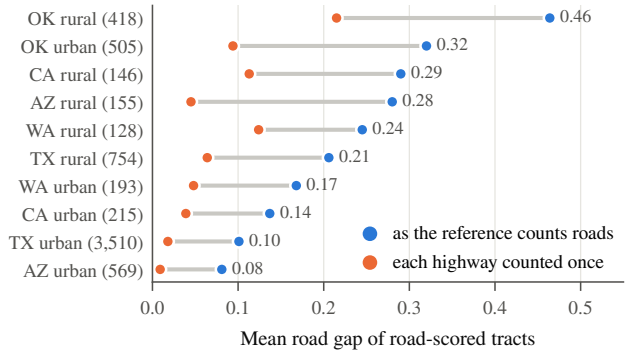


Figure 2: Bookkeeping makes up most of the road gap in every region and urbanity, far more in Oklahoma than in Arizona or Texas (tracts per row in brackets; §3).

Part A, Best Bias Discovery: where each criterion is answered

Novelty §2, §4.1: a pattern outside the grid that bends the grid. **Evidence** §3: lengths, tests on the hidden scores, robustness, named tracts, a worked example. **Impact** §4: understated disadvantage, a distorted worklist, evacuation, Census and FEMA data (§4.5). **Reproducibility** §3.8: 33 checks in a second; Appendix A. **Also:** limits §5, fix §6, prior work §7.

Part B, Best Documentation: the organisers' checklist

Overview: §8. **Data sources:** §9. **How each component was computed:** §10. **Edge cases** (zero-population, water-dominated, data-sparse): §11, Table 10. **Alternative weightings and why:** §12, Table 11. **Reproducibility:** §13, Appendix A; both finals rebuild from the package to 3×10^{-16}.

1 Introduction

The challenge scores each census tract by how much of a reference map Overture lacks. For roads, it compares the length of Overture's highway-class roads inside the tract with the length of primary and secondary road in Census TIGER/Line (codes S1100 and S1200 [3]). The organisers' scorecard then compares these gaps across urban and rural tracts, tribal land, social and climate vulnerability and hazard exposure [1].

We show that the road part of that score mostly measures how the reference is counted, not how much of the highway network is missing, and that this distorts the very comparisons the scorecard exists to make. Our contributions:

1. **The size of the bookkeeping**: two counting rules from public challenge discussion (§7), measured here from the package alone and tested against the hidden scores, make up 72% of the mean road gap (55–80% under any weighting), from 0.07 in urban Arizona to 0.23 in urban Oklahoma (§2–3).
2. **The scorecard it bends**: removing it raises the rural and wildfire-hazard ratios by 17% and 9% and removes nearly all of the social-vulnerability disparity (§4.1).
3. **The places it hides**: a worst-tenth worklist holds 321 tracts whose highways are mapped and leaves out 321 that are mostly rural, tribal and fire-exposed; we name examples on both sides (§4), and public Census and FEMA data show the hidden places are older, less connected and at higher risk (§4.5).
4. **A check anyone can run**: 33 of the paper's numbers recomputed in under a second from a few megabytes of the package, and a label-free method whose finals rebuild to within 3×10^{-16} (§3.8, Part B).

Part A Best Bias Discovery

2 The pattern the scorecard does not surface

In one sentence: the road score counts a highway on a tract boundary in both tracts, and counts a highway once per route number and per name, so much of what it reports as missing road is bookkeeping.

The organisers' formula cuts both road layers with the tract polygons by an overlay intersection [2]. We call the reference length a tract receives this way its *bookkept reference length*, and split it into three parts that can be measured from the package alone:

1. **Interior highway**: highway that runs through the tract, measured as the length of Overture road inside the tract that carries an Interstate, US or state route number.
2. **Boundary highway**: highway whose centreline is the tract boundary. Census tract boundaries follow visible features [4], most often roads, so the boundary polyline in the package's tract file is the reference's own highway line, and the overlay counts it inside both tracts. We sample every tract boundary every 10 m and match each sample to the nearest Overture route within a region-specific distance (Table 1); a stretch is counted when Overture's copy lies outside the tract.
3. **Duplicate records**: extra records on the same stretch. The reference holds one record per route number and per name, so where two routes share a road, or a route also carries a local street name or an honorific highway name, the same metres are counted again. We count the extra route numbers and names on each stretch from Overture's own route and name attributes.

Table 1: What the reference counts, by region. The distance cut is the furthest a boundary sample may sit from Overture's highway and still be read as that highway, fitted per region by maximum likelihood from the boundary-to-highway distances, with no reference to the score. Interior: route-numbered highway inside tracts. Boundary: boundary highway whose Overture copy lies outside the tract. Extra records: length of additional route numbers and names.

Region	Road-scored tracts	Cut (m)	Interior (km)	Boundary (km)	Extra records (km)
Oklahoma	923	21.4	24,293	2,849	20,095
Texas	4,264	19.2	51,710	9,810	30,168
Arizona	724	15.8	11,422	2,288	13,805
Washington	321	11.3	8,142	832	3,741
California	361	10.6	6,984	980	4,127

A tract follows the pattern when parts 2 and 3 are a large share of its reference length, or when they change its road gap by a large amount.

2.1 Why the scorecard cannot see it

The automated scorecard takes each tract's road gap as given and compares averages across fixed groups: urbanity, social and climate vulnerability, tribal land and hazards. It never asks what the reference length inside a tract consists of. A boundary highway counted on both sides, and a stretch counted once per name, look to it exactly like highway that Overture is missing. The pattern is a data-source behaviour of the reference, not of Overture, and it depends on region and on where a tract's highway runs relative to its boundary, so no split by the scorecard's strata can separate it from real coverage. It is the kind of discovery the challenge describes as "a data source behavior". Worse, because the bookkeeping is uneven across places, it shifts the scorecard's own ratios: it is a pattern outside the grid that bends the grid (Section 4.1).

3 The evidence

3.1 Bookkeeping is almost half of the reference length

Over all tracts, the package shows 102,551 km of route-numbered highway inside tracts, 16,759 km of boundary highway whose Overture copy lies outside the tract, and 71,937 km of extra route and name records (Table 1). Overture's side also contains 12,448 km of motorway ramps that the reference's road classes leave out. At face value, boundary highway and duplicate records are 46% of the bookkept reference length.

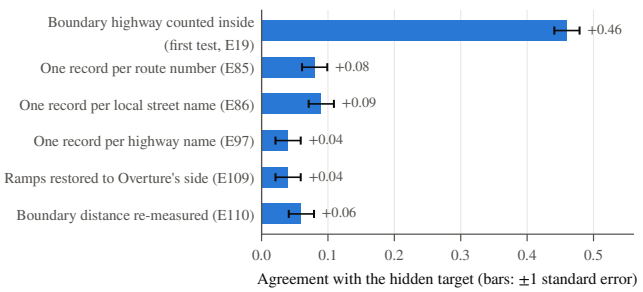


Figure 3: Each counting rule's agreement with the hidden target, read from the change in public score when it was added alone. Bars are ±1 standard error of the public sample.

The distance cut itself differs by region. In Oklahoma and Texas the gap between the tract line and Overture's highway has a long tail and the cut sits near 20 m; in California and Washington it is near 11 m. So the same highway geometry is matched differently in different regions, and that is visible in the package without any model.

3.2 Each counting rule was tested against the hidden scores

We do not have to assume that the reference behaves this way. Each rule was added to our model as a single change and submitted once, and the change in the public score measures how strongly the change agrees with the hidden target (Section 12 gives the arithmetic; zero means no agreement, and one standard error is about 0.02). Every rule raised the score, by 2 to 24 standard errors (Figure 3).

The boundary rule is the largest single effect we measured in the whole competition. When we later fitted the model's weights to everything the leaderboard had revealed, the boundary highway kept a weight of 0.92 and each extra record about 0.53 to 0.55: the reference counts boundary highways almost fully, and each duplicate name about half the time. Every number in the rest of Part A uses these fitted weights, not the face-value counts.

3.3 Bookkeeping adds 0.07 to 0.23 to the road gap, unevenly by region

To turn lengths into a road gap we use our calibrated simulator (Section 10): it draws the hidden reference for each tract many times and averages the resulting gap. Its road part is the one tested in Figure 3. We run it twice with the same calibration: once as the reference counts, and once with the boundary and duplicate-record terms switched off, so each highway counts once, in the tract it runs through. The difference is the road gap that the bookkeeping adds (Figure 2, Table 2).

The comparison is within region and within urbanity, so it is not the urban/rural or regional split that the scorecard already shows. The bookkeeping adds 0.23 in urban Oklahoma (95% bootstrap interval 0.21–0.24), 0.12 in urban Washington, 0.10 in urban California, 0.08 in urban Texas (0.079–0.088) and 0.07 in urban Arizona (0.06–0.08). Rural tracts show the same ordering. Of the two parts, the duplicates add more in every region; the boundary part is largest in rural Arizona (0.08 on its own). Figure 4 shows where it falls inside Oklahoma, the region where it is largest.

Table 2: Mean road gap of road-scored tracts (0: the map has all of the reference's highway; 1: none). Urban and rural follow the Rural-Urban Commuting Area code (rural is code 4 or above).

	Urban		Rural	
Region	as scored	counted once	as scored	counted once
Oklahoma	0.32	0.09	0.46	0.22
Arizona	0.08	0.01	0.28	0.05
Texas	0.10	0.02	0.21	0.06
California	0.14	0.04	0.29	0.11
Washington	0.17	0.05	0.25	0.12

Table 3: Share of the mean road gap of road-scored tracts that is bookkeeping, under different counting weights (same calibration; counted once, the mean road gap is 0.047 in every row).

Weights	Boundary	Duplicates	Share
Fitted (used in this paper)	0.92	×1.00	**72%**
Every record in full	1.00	×1.81	80%
Half of the fitted	0.46	×0.50	55%
Duplicate rule alone	0	×1.00	62%
Boundary rule alone	0.92	0	46%

3.4 The headline does not depend on the fitted weights

The 72% uses the weights the hidden scores chose (boundary 0.92, duplicate records about 0.54). Table 3 runs the same calibrated simulator with other weights. Counting every record in full gives 80%; halving both weights still gives 55%; the duplicate rule alone gives 62% and the boundary rule alone 46%. So "most of the road gap" holds under every weighting except one rule on its own, and the ordering of regions does not change (`writeup_numbers.py sensitivity`).

3.5 It changes which tracts look worst

Ranked by road gap, 660 tracts form the worst tenth. With each highway counted once, 321 of them (49%) fall out of that tenth (Figure 5). They hold 507,345 homes and 1.19 million people. Over the whole map, 1,548 tracts get at least 0.2 of their road gap from bookkeeping; they hold 2.5 million homes and 5.8 million people.

3.6 Ten named tracts where it is nearly all of the gap

Table 4 lists, for each region, tracts where the bookkeeping makes most of the road gap; `docs/writeup/tracts.csv` gives them with every measured length.

Table 4: Tracts where the bookkeeping makes most of the road gap, two or three per region. Homes are ACS housing units from the package.

Tract (GEOID)	Place	Homes	Road gap as scored	Counted once
06101050503	Yuba City, Sutter County, CA	2,040	0.76	0.00
06113010902	Woodland, Yolo County, CA	2,372	0.83	0.01
48451000200	San Angelo, Tom Green County, TX	1,840	0.86	0.01
48215020505	McAllen, Hidalgo County, TX	981	0.84	0.00
48113019025	Dallas, Dallas County, TX	1,503	0.83	0.00
04025000209	Yavapai County, AZ	2,790	0.87	0.00
40017301201	Canadian County, OK (Cheyenne and Arapaho tribal statistical area)	1,959	0.86	0.02
40131050303	Rogers County, OK (Cherokee tribal statistical area)	1,194	0.84	0.01
53063012701	Spokane, Spokane County, WA	1,897	0.72	0.01
53075000400	Whitman County, WA	2,329	0.86	0.23

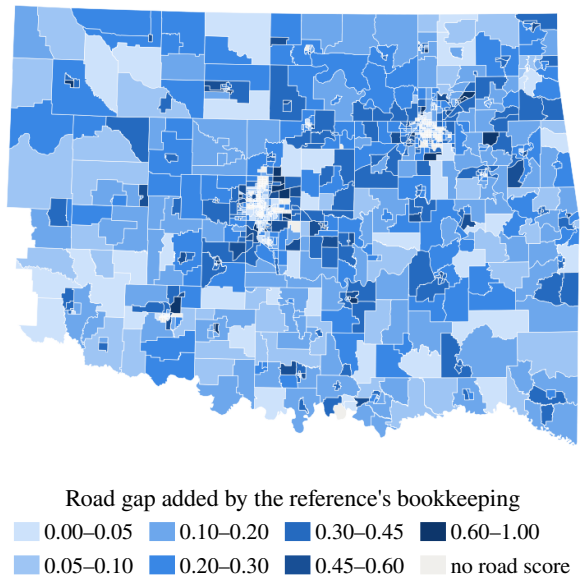


Figure 4: Road gap added by the reference's bookkeeping, eastern Oklahoma. It is largest in the Oklahoma City and Tulsa metro tracts, where highways run on tract lines and carry several route numbers and names.

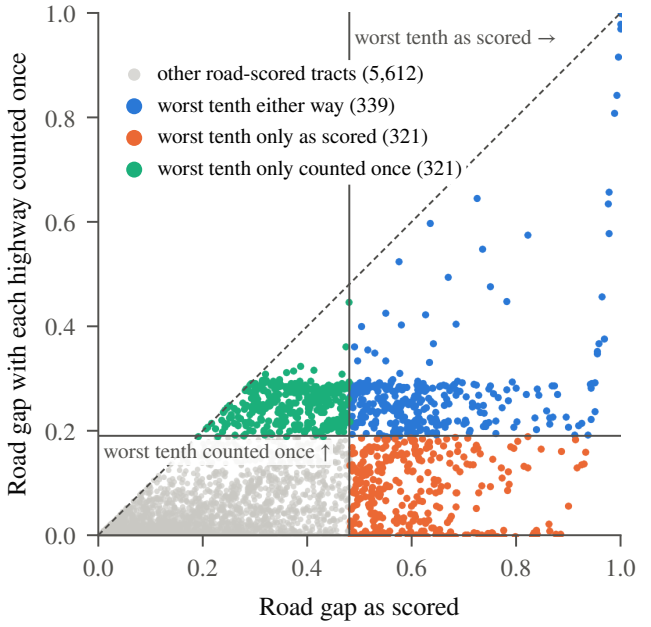


Figure 5: Every road-scored tract, as scored against counted once. Orange tracts are in the worst tenth only because of the bookkeeping: 321 of the 660 worst-scored tracts.

3.7 One tract, end to end

Census tract 505.03 sits on the eastern edge of Yuba City in Sutter County, California, with 2,040 homes. State Route 99 runs down its whole eastern side, and the tract line there is the highway's centreline (Figure 6). Overture maps State Route 99 along all 1,644 m of that edge, a median 2.8 m east of the tract line (0.4 to 3.8 m over 90% of its length). None of it falls inside the tract. Inside the tract, Overture's only highway-class road is 452 m of Franklin Road along the northern edge. So the score compares about 0.45 km of mapped road with 1.6 km of reference highway, and our simulator puts the tract's road gap at 0.76. With State Route 99 counted once, in the tract east of the line where Overture draws it, the gap is 0.00. Nothing about the map of this tract would need fixing.

3.8 Reproducing Part A

Part A can be checked in two ways: quickly from our published per-tract results, or from scratch.

In under a minute. We publish every tract's road gap, as scored and counted once, as `tract_results.csv` in `docs/writeup/`. With only the package's strata tables (a few megabytes), one command checks 33 of the paper's numbers against it, our final submission, the package and the Census and FEMA extract, in under a second, and can remeasure the worked example from Overture's roads:

- `python src/download.py --strata` then `python src/quick_check.py`; add `--geometry` after `python src/download.py northern-ca` for the Yuba City measurement. It prints "33 of 33 checks match the paper" (34 of 34 with the measurement).

Table 5: The organisers' scorecard on our final submission (E136), as scored and with each highway counted once. Ratios are the mean score of the first group over the second; median splits as in the scorecard.

Scorecard ratio	As scored	Counted once	Change
Rural vs urban	2.34	**2.73**	+17%
Wildfire hazard, high vs low	1.67	**1.81**	+9%
Summer heat, high vs low	0.59	0.64	+10%
Winter drought, high vs low	1.30	1.32	+2%
Tribal vs non-tribal	2.77	2.59	−7%
Climate vulnerability, high vs low	1.40	1.27	−9%
Social vulnerability, high vs low	1.16	**1.01**	−13%
Road gap alone: rural vs urban	1.67	**3.32**	+99%
Road gap alone: tribal vs non-tribal	2.98	**5.23**	+75%

From scratch. With the full package and the per-tract features built (Appendix A), the simulator's outputs themselves are recomputed in about five minutes: `python src/writeup_numbers.py` (with `displaced`, `scorecard` and `sensitivity` for Tables 6, 5 and 3), then `python src/external_context.py` for Section 4.5 (add `fetch` to download its public data again) and `python src/paper_figures.py` for the figures.

4 Who is affected, and why it matters

In one sentence: the bookkeeping understates how far rural and fire-exposed places fall behind, invents most of the reported gap for socially vulnerable ones, and sends a worklist to mapped highways in towns and cities while rural, tribal and fire-exposed tracts wait.

4.1 It bends the scorecard built to reveal inequity

The scorecard exists to show which people the map serves worse: it divides the mean score of rural, tribal, vulnerable or hazard-exposed tracts by that of the rest. Table 5 and Figure 1 compute it on our final submission twice: as scored, which reproduces the card the leaderboard prints for that file, and with each highway counted once (a road-scored tract's score loses the road gap the bookkeeping adds, divided by its three components). The bookkeeping is heaviest in towns and cities, where highways run on tract lines and carry several route numbers, so it lifts urban scores and flattens the comparisons. Counted once, rural tracts are 2.73 times as far behind as urban ones, not 2.34; on the road gap alone the ratio doubles, from 1.67 to 3.32, and the tribal one rises from 2.98 to 5.23. The wildfire-hazard ratio rises from 1.67 to 1.81. In the other direction, the scorecard's social-vulnerability disparity (1.16) almost vanishes (1.01): nearly all of it is bookkeeping in vulnerable urban tracts, not missing map. A reader of the official card would underrate the disadvantage of exactly the rural and fire-exposed places that have the fewest alternatives when a road is wrong, and would chase a social-vulnerability gap that is mostly an artefact.

Table 6: The two groups the bookkeeping swaps on a worst-tenth worklist. *Hidden*: in the worst tenth only when each highway is counted once. *Mapped*: in it only as scored. Wildfire hazard is the package's USFS wildfire hazard potential; burned is MTBS, 1984–2023; bookkeeping share is boundary plus duplicate length over the reference length. Source: `numbers.json`, key `displaced`.

	Hidden	Mapped	All road-scored
Tracts	321	321	6,593
People (millions)	1.00	1.19	27.1
Rural (RUCA ≥ 4)	**71%**	38%	24%
On tribal land	**58%**	26%	11%
Wildfire hazard above median	**83%**	61%	54%
Burned since 1984	**50%**	11%	13%
Bookkeeping share (median)	29%	75%	57%

4.2 A worklist built from the score points at mapped highways

Road-coverage scores are the kind of number an emergency-management office or a mapping programme uses to decide where to check evacuation routes, and where to add or verify highways before a fire, flood or hurricane season. Take the 660 tracts with the worst road gap as that worklist. Half of it (321 tracts, 1.19 million people) is there because of counting rules, and for those tracts the work would find nothing to fix: the highway is on the map, a few metres from the line. Over the whole map, bookkeeping adds at least 0.2 to the road gap of 1,548 tracts, home to 5.8 million people, from the Cheyenne and Arapaho and Cherokee tribal statistical areas of Oklahoma to Spokane, Yuba City, Woodland, San Angelo, McAllen and Dallas. What they share is geometry: a highway on the tract's edge, or one that carries several route numbers or names.

4.3 Whose gaps it hides: rural, tribal and fire-exposed tracts

Removing the bookkeeping does not only shrink gaps; it changes whose gaps they are. The 321 tracts that join the worst tenth when each highway is counted once have a gap that is, by the score's own definition, mostly coverage: bookkeeping is a median 29% of their reference length, against 75% in the tracts they would replace. Table 6 compares the two groups. The hidden tracts are 71% rural, 58% on tribal land and 83% above the median wildfire hazard potential, and half have burned since 1984. The tracts that crowd them out are 38%, 26%, 61% and 11%. So the score as it stands ranks the places where a missing evacuation route is hardest to work around below places whose highways are already on the map.

4.4 What it means in named places

Yuba City, California (mapped, but flagged). Tract 505.03 sits on State Route 99. In February 2017 the Oroville spillway emergency led to the evacuation of about 188,000 people from Butte, Yuba and Sutter counties [9]. The Sutter County grand jury then called for planned evacuation routes in a

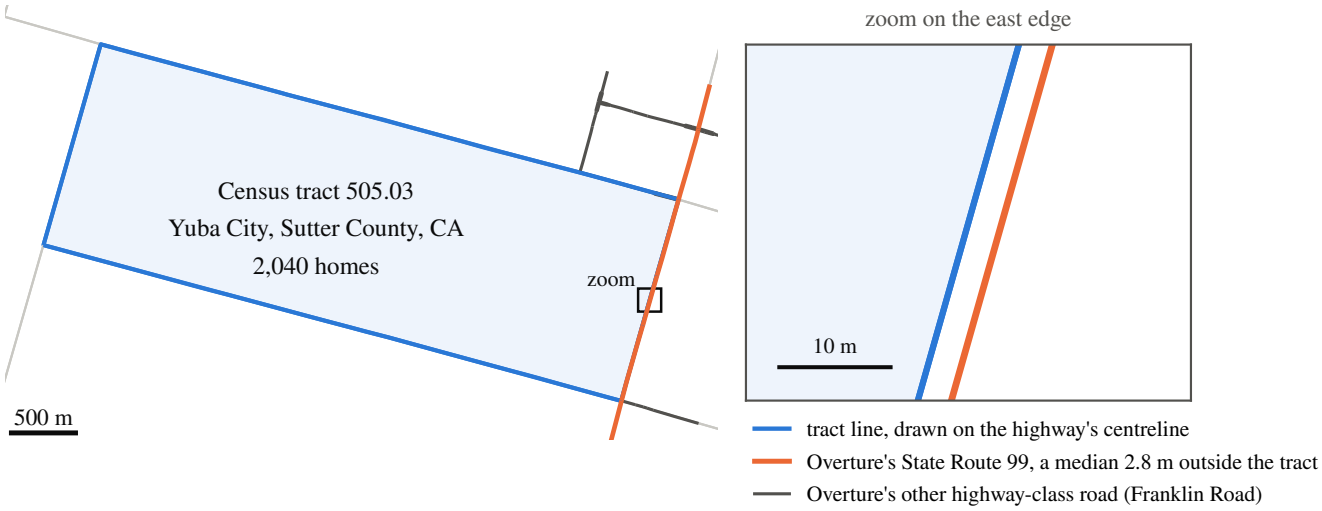


Figure 6: Yuba City tract 505.03. The tract line and Overture's State Route 99 run side by side, a median 2.8 m apart, so the reference counts a highway inside the tract that Overture places next door.

Table 7: Six hidden tracts: rank by road gap among the 6,593 road-scored tracts, as scored and with each highway counted once (the worst tenth is ranks 1–660). All six have burned since 1984.

Tract (GEOID)	Where (tribal area)	People	Scored → once
40085094300	Love Co., OK (Chickasaw)	2,023	665 → 30
40021977600	Cherokee Co., OK (Cherokee)	5,307	689 → 40
40029388200	Coal Co., OK (Choctaw)	1,702	1,063 → 48
48007950101	Aransas Co., TX	1,300	949 → 49
06103000100	Tehama Co., CA	4,670	1,139 → 50
40069660198	Johnston Co., OK (Chickasaw)	2,906	1,200 → 51

region with few ways out, and found that opening a second lane on Highway 20 or 99 was seen as unrealistic [10]. So Route 99 is a road the region's evacuation planning depends on. The score marks this tract as missing three quarters of its highway, yet Overture carries all 1,644 m of Route 99 along its edge (Figure 6). An agency checking evacuation corridors by the score would spend its first visit on a road that is already mapped.

Aransas County, Texas (hidden). Hurricane Harvey's centre came ashore in Aransas County in August 2017, near Rockport and again west of Holiday Beach [11]. Tract 9501.01 there ranks 949th by road gap as scored and 49th with each highway counted once: its gap is coverage, not counting, yet it sits outside a worst-tenth list.

Tribal Oklahoma and wildfire country (hidden). Four of the six tracts in Table 7 lie in the Chickasaw, Cherokee and Choctaw tribal statistical areas; all four have burned since 1984 and sit at or above the 70th percentile of wildfire hazard among the challenge's tracts. The Tehama County tract is at the 99th percentile. These are the places where a single missing or misclassified highway can decide how a community leaves ahead of a fire, and the score as it stands ranks them below tracts whose highways are mapped.

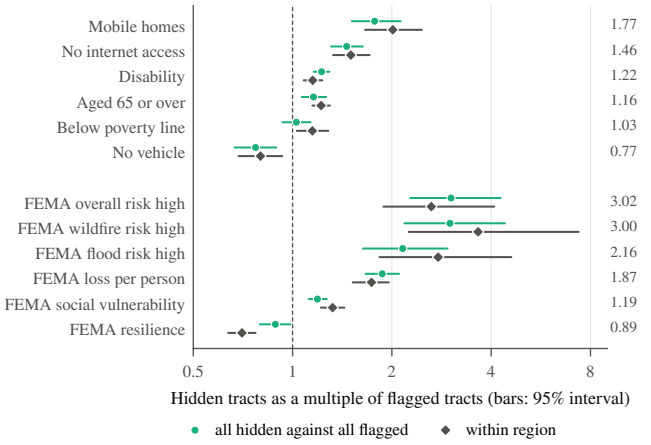


Figure 7: Hidden tracts against flagged tracts on all twelve measures; 1 means no difference, and the numbers on the right are the pooled ratios. *Within region* compares the groups inside each region, weighted by where the hidden tracts are. Resilience is the one measure where lower means worse off.

Comparisons between places. A state agency comparing Oklahoma's urban road gap (0.32) with Arizona's (0.08) would conclude that Oklahoma's highway map is four times less complete. Counted once, the two are 0.09 and 0.01. Most of the difference is how often Oklahoma's highways run on tract lines and carry several names, not how well they are mapped.

4.5 What public data outside the package say

The prize form lets a discovery draw on public data beyond the package, so we asked two independent sources who lives in the swapped tracts: the Census American Community Survey 2020–2024 [12] and FEMA's National Risk Index v1.20 [13]. The package holds composite vulnerability indices but not these measures, and no scored file uses them (Appendix B). We fixed twelve measures before computing any comparison, and we report all twelve (Table 8, Figure 7).

Table 8: The two groups of Table 6 on public data outside the package. Shares are pooled over each group's tracts; *all* is every road-scored tract. FEMA *high* means rated Relatively High or Very High. A higher resilience score is better.

Measure	Hidden	Flagged	All
Homes that are mobile homes	16.5%	9.3%	8.2%
Households, no internet access	11.3%	7.8%	6.8%
People with a disability	18.8%	15.3%	13.6%
People aged 65 or over	20.1%	17.4%	15.0%
People below the poverty line	15.3%	14.9%	13.9%
Households, no vehicle	4.1%	5.3%	5.5%
FEMA overall risk high, tracts	41.4%	13.7%	20.9%
FEMA wildfire risk high, tracts	33.6%	11.2%	10.7%
FEMA inland flood risk high, tracts	34.3%	15.9%	26.4%
FEMA annual loss per person	$843	$451	$489
FEMA social vulnerability score	69.3	58.2	55.7
FEMA community resilience score	27.0	30.5	30.1

Ten of the twelve point the same way, and all ten hold within regions. The hidden tracts have nearly twice the share of mobile homes of the flagged tracts (16.5% against 9.3%), and more people aged 65 or over, more people with a disability and more households with no internet access. FEMA rates three times as many of them high for wildfire risk (34% against 11%) and for overall risk (41% against 14%), and twice as many for inland flooding; it scores their community resilience lower and expects nearly twice the annual loss per person ($843 against $451). Two measures do not follow the pattern. Hidden tracts have fewer households with no vehicle (4.1% against 5.3%), so the bookkeeping does not hide car-free households. Poverty is level overall (15.3% against 14.9%), and higher in hidden tracts only within regions.

In absolute terms the hidden tracts hold about 78,000 mobile homes, 189,000 people with a disability and 44,000 households with no internet access. Two of these needs were named after Oroville: the Sutter County grand jury asked for plans for people with disabilities and recorded concern that not everyone has internet access for emergency information [10].

The bookkeeping also blunts the score as a guide to risk. Of the 706 road-scored tracts FEMA rates high for wildfire risk, a worst-tenth worklist holds 110 as scored and 182 with each highway counted once; of the 1,742 rated high for inland flooding, 139 and 198; of the 1,376 rated high overall, 130 and 219. The rank correlation between a tract's road gap and FEMA's wildfire risk score rises from 0.33 to 0.45 (0.31 to 0.45 within regions). FEMA's index is built without this road score, so this is an outside sign that taking the bookkeeping out points the road gap at the places with more to lose.

5 Limits

The split between bookkeeping and real coverage comes from our calibrated simulator, not from the hidden reference itself, which is not in the package. The simulator's road part was tested against the hidden scores (Figure 3), and Table 3 shows how far other weights move the share (55–80%, or 46% for the boundary rule alone). The scorecard comparison of Table 5 uses our final submission as the best available estimate of each tract's score, and removes the bookkeeping from a road-scored tract's score as one third of its added road gap, which assumes the building and places components are defined (they are in almost every road-scored tract). The leaderboard tests use the public 30% of tracts, so their agreement values carry that sample's noise; the private 70%, revealed after the close, backs the two rules that can be read directly from it (the boundary rule took 0.08296 to 0.07708 from E18 to E19, and the re-measured boundary distance 0.05790 to 0.05747 from E109 to E110). The hidden tracts of Table 6 are those where the gap is *not* bookkeeping; their remaining gap rests largely on each region's calibrated ratio between the reference's road classes and Overture's, so the list says where to look, not that a particular road is missing on the ground. And we cannot say how a given agency actually uses the score; the worklist in Section 4 illustrates a use the score invites. Section 4.5 describes groups of tracts, not people: Census figures are survey estimates, and FEMA describes its index as a nationwide planning tool, not a substitute for local risk assessment. It shows who lives in the hidden tracts, not that their highways are missing. The government reports cited in Section 4 give context only; no computed result in this paper rests on them.

6 What would fix it

Three changes to the reference side would remove most of the effect: count a highway that lies on a tract boundary once, on the side where it is mapped (or half on each side); remove duplicate records of the same geometry before measuring length; and give both layers the same treatment of ramps (both include them, or neither does).

7 Prior work

Both counting mechanisms were first described by other participants, and we credit them here. A participant's post "The road gap mostly measures TIGER's data model" [6] makes the general claim first. Another participant's entry "Counted twice: the score charges tracts for highways that run along their edge" [7] describes the boundary mechanism and reports that modelling it improved their public score. A third participant's public method notes [8] state the duplicate-record mechanism (the reference keeps one record per edge and name, so a road carrying two route numbers or a local street name is counted again) and fit it on the leaderboard. Our contribution is to measure both from the package alone and test each against the hidden scores (extending the record rule to honorific highway names, E97); to show that together they make up 72% of the mean road gap, with duplicate records

the larger part in every region; to show that the share differs by region; and to trace the consequence for people: the rural, tribal and fire-exposed tracts the bookkeeping pushes down a worklist, with named tracts and a worked example.

Part B Best Documentation: methodology

The method in five plain steps

1. **Measure the map.** For every tract, lengths and counts from Overture's roads, places and buildings, and the package's tables.
2. **Imagine the hidden reference.** A simulator draws, many times, how much reference road, how many facilities and how many businesses each tract could have, applies the organisers' formula, and averages.
3. **Ask the leaderboard one question at a time.** Each idea goes up once as a small change to a scored file; two scores give its agreement with the hidden target exactly.
4. **Combine without overfitting.** Package-only pieces are fitted to the shape of the best combination of scored files, never to a score directly, and regularised for the private split.
5. **Correct what is left.** A seeded learner predicts the simulator's remaining error from the package; its unmeasured part is the last step. Every final rebuilds from the package to 3×10^{-16}.

8 Overview: predicting a score with no labels

The challenge asks for a coverage-gap score for each of 9,794 census tracts in five regions (eastern Oklahoma, eastern Washington, the Maricopa region of Arizona, northern California and south-central Texas). No tract-level labels exist. The score compares Overture's roads, buildings and places with reference layers (TIGER/Line roads, Microsoft building footprints, USGS/HIFLD facilities and County Business Patterns) that were removed from the package and that the rules forbid us to use. So our model has to estimate, for each tract, how much reference road, how many reference buildings and how many reference facilities and businesses there are, from what the package does contain, and then apply the organisers' formula.

We did this with a simulator: for each tract it draws the hidden reference many times from distributions built on Overture's own data, applies the scoring formula to each draw and averages the result. The simulator is calibrated to published averages. Each new idea about how the reference behaves was added to our best file as a small change and submitted once; the change in score measured how much the idea agrees with the hidden target. The final files combine the measured ideas (Figure 8). Our selected finals scored 0.05556589 and 0.055506023 on the public split and 0.056132715 and 0.05613082 on the private one; for scale, predicting the mean everywhere scores 0.0890. Figure 9 shows how we got there.

9 Data sources

The scored submissions use only the challenge package [2] (Table 9). External data appears nowhere in them. Part A goes outside the package once, in Section 4.5, as the prize form allows, with the two public datasets listed in Appendix B, and it cites three public reports for context.

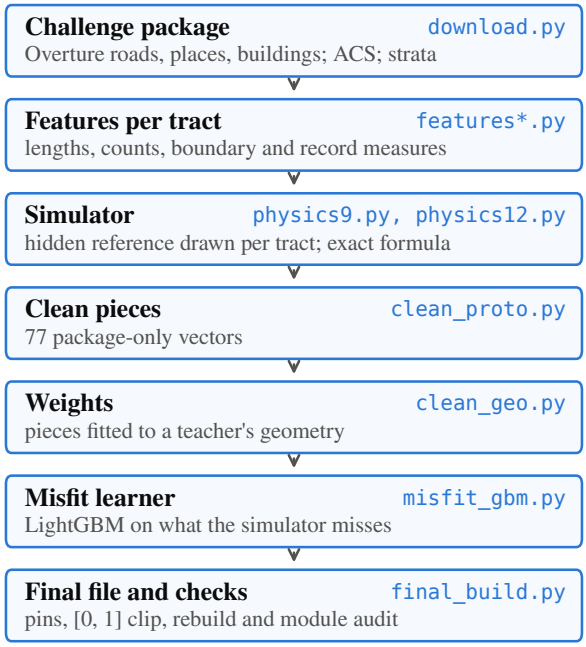


Figure 8: From the package to the final file; each box names the script that does it.

10 How each component was computed

The score is the mean of the defined components; a component is undefined where the tract has no reference of that kind. Each component is

$$\text{gap} = 1 - \min\left(1,\ \frac{\text{Overture amount}}{\text{reference amount}}\right).$$

We compute the Overture amount exactly as the organisers describe it and simulate the reference amount.

Roads. The Overture amount is the length of motorway, trunk, primary and secondary road inside the tract, filtered on class only, as the organisers' text says (this keeps motorway ramps in). The reference amount is simulated as a per-region scale times the bookkept reference length of Section 2 (interior highway, plus boundary highway, plus duplicate records, minus Arizona's route-log highway names, which are not reference records), with a per-tract log-normal spread. The road component is defined where the bookkept length is positive; this reproduces the organisers' published counts of road-undefined tracts to within 2% per region (Arizona's exactly). The per-region scales are calibrated to a competitor's published mean scores for 16 region, urbanity and burn cells (a published aggregate, listed in `src/published.py`).

Facilities half of the places component. For fire stations, ambulance services and schools the Overture amount is the count of places whose primary category is `fire_department`, `ambulance_and_ems_services` or one of six school categories. The reference amount is the Overture count plus stations or schools the reference has and Overture does not list under those categories: places filed under another category but named like a station or school, stations traced only as buildings, one record per fire agency, and

Table 9: Package layers used, and what we take from each.

Layer	What we take from it
Overture roads, release 2026-08-19.0 [5]	road length by class inside each tract; route numbers and names; ramps
Overture places	counts by category; names, confidence and source metadata
Overture buildings	footprint counts and sizes
Overture rail, infrastructure	rail lines (to set railway boundaries aside); transit stops
ACS housing (B25001)	housing units per tract
Tract polygons (TIGER 2025)	geometry, land area, population
Strata tables	urbanity (RUCA), tribal land, SVI, CVI, burned area, heat and drought

a background rate in tracts with no trace at all. These rates are calibrated to two published counts (740 fire-station blackouts, 270 of them misfiled) and to the published fire-to-school gap ratio.

Business half of the places component. The Overture amount is all places in the tract. The reference amount (County Business Patterns establishments) is simulated from building counts, Overture's own record metadata and a wide log-normal spread, scaled so the places component matches its published average of 0.052.

Buildings. The building gap is almost always near zero (Overture's footprints largely come from the same machine-learned source as the reference), so it is a constant of 0.0065.

The simulator's mechanism weights (how fully boundary highways and duplicate names count, the rural station rate and so on) were fitted to everything our scored files had revealed about the hidden target, then its prediction in the directions no file had measured was submitted as a bet: a method we took from a participant's public notes [8]. Its code is `src/physics9.py` (with `src/physics12.py` for the refits of 31 October); the fitted weights are committed in `params/`.

11 Edge cases

Table 10 lists every edge case the organisers name, and the rest we met. Two further rules keep files valid: every file has 9,794 rows in the sample submission's order, with 11-character GEOIDs kept as text; and our writer refuses to overwrite a file that has already been scored, because its score is the only measurement of that exact vector.

Table 10: Edge cases and how each is handled.

Case	Tracts	Handling
Zero population	27 (82 with no homes)	scored like any other tract; every ratio with a population or housing denominator is guarded
Water-dominated (zero land area)	2	no Overture road, building or place, so every defined component is $1-\min(1,0/\text{ref})=1$; set to exactly 1.0, later confirmed by a score change
Data-sparse: no Overture data at all	the same 2	as above
Data-sparse: no Overture place of a type	71% fire, 93% EMS, 21% school	the simulator draws whether the reference holds one: gap 1 if it does, component undefined if not
Road component undefined	3,201	score is the mean of the places and building components
Eastern Washington	415	no published calibration; its road scale is the median of the other four regions'
Values outside $[0,1]$	any	clipped to $[0,1]$ after combining

12 How the final weights were set, and the alternatives we tried

Measuring an idea with one submission. Each idea about the hidden reference was submitted once, as a small dose d on top of a scored base b. Because the score is an RMSE, two scores give the covariance of the change with the target exactly: $\overline{d\,(y-b)} = \frac{1}{2}\left(\overline{d^2} + \text{MSE}_b - \text{MSE}_{b+d}\right)$, where the bars are means over the public tracts. Scaled by the dose's size, this is the idea's *agreement* with the target (Figure 3); its standard error over the 30% public sample is about 0.02.

Building the finals. The finals are built in three steps. First, 77 package-only pieces (the simulator's variants and the raw vectors of every idea we tested, rebuilt from the package) are fitted, by least squares and without using any score, to reproduce the shape of a *teacher*: a weighted combination of our scored files. Second, measured ideas are added on top at their measured strength. Weighting by geometry rather than by score matters: when we let the scores choose the pieces' weights directly, the fit exploited the noise of the 30% public sample and the file scored 0.0016 worse than predicted (E123 in Table 11).

The misfit learner. Third, a learned correction. The part of the target the scored files already reveal is known exactly (the teacher), and so is the simulator's projection on the same files; their difference is what the simulator gets wrong, as far as the scores can see. A seeded LightGBM model (`src/misfit_gbm.py`) learns that difference from about 400 package features, out of fold (R^2 0.45; the facility features matter most). Its prediction in the directions no scored

Table 11: Weightings and ideas tried in the last three days, and why each was kept or dropped. Scores are RMSE; every decision used the public score, and private scores were revealed after the close (`submissions/ledger.csv` lists every scored file; `docs/HISTORY.md` tells the whole story).

Weighting or idea	Public	Private	Kept?
Pieces weighted by their measured agreement (E123)	0.05702	0.05773	no: fitted the public sample's noise
Pieces fitted to our best earlier file's geometry (E124, 63 pieces)	0.05631	0.05624	superseded
Same with 77 pieces (E125)	0.05582	0.05637	superseded
Plus the fitted simulator's new part (E127)	0.05576	0.05616	yes, as a piece
Pieces fitted to a combination of 69 scored files (E129)	0.05559	0.05617	superseded
Same teacher, light regularisation (E131)	0.05557	0.05615	base of final B
Same teacher, stronger regularisation (E132)	0.05563	0.05616	base of final A
Housing term in the business reference (E130)	0.05590	0.05629	no: disagreed with the target
Vulnerability term in the business reference (E133)	0.05567	0.05627	no: carried no signal
Misfit learner's step on E132 (E134)	0.05557	0.05613	**final A**
Learner v2, beyond E134 (E135)	0.05561	0.05610	no: nothing new on public
E131 plus the learner's step at its optimum (E136)	0.05551	0.05613	**final B**
Simulator refit on all scored files (E137)	0.05552	0.05623	no: too small

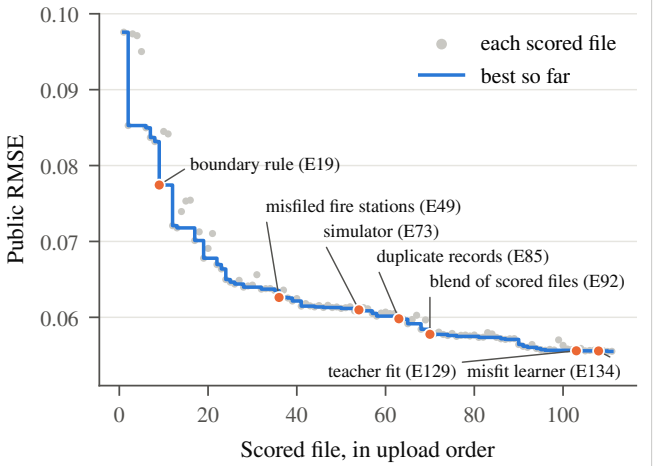


Figure 9: Public RMSE of every scored file, in upload order, with the discoveries that moved it most. Both finals scored 0.05613 on the private split (Table 11). Early exploratory files that we later excluded (Section 13.1) are included, so the curve is the history, not the finals.

Table 12: Predicted and actual public RMSE. We predicted public scores only; private scores are in Table 11.

File	Predicted	Actual
E125	0.05590	0.05582
E129	0.055590	0.055588
E131	0.05553	0.05557
E132	0.05565	0.05563
E136	0.055509	0.055506

file has measured is the step: E134 added it to E132 and read an agreement of +0.030, the last idea that paid. A better-fitting second version (R^2 0.58) added nothing beyond it on the public score (E135), though it later proved our best private file, and a refit of the simulator on all 104 scored files read +0.010 (E137), too small to use, and scored worse than E136 on the private split (0.05623 against 0.05613).

Why two finals. Weights read from the public 30% fit some of its noise, so a file tuned harder to the public split does a little worse on the private 70%. We simulated this with synthetic targets and the same file structure: a teacher regularised more strongly gives up a little on the public split (0.00006 between E131 and E132) and should gain about 0.0001 to 0.0002 on the private one. We selected one file of each kind, each with the learner's step: final A is E132 plus the step (E134), final B is E131 plus the step at its measured optimum (E136). The private scores did not show that gain: E132 scored 0.056156 against E131's 0.056149, and the two finals finished level at 0.056133 and 0.056131. The learner's step did hold on the private split, taking E132 from 0.056156 to 0.056133. Our best private file was the learner's second version, E135, at 0.056099, which we had set aside on its public score.

12.1 Validation: predicting a score before uploading it

For a file that is a combination of scored files, the score can be predicted before uploading it from the files' own scores and their second moments, by the identity above. We used the prediction to decide whether a file was worth a submission (Table 12). The predictions were within 0.0001, and the files did at least as well as predicted when the pieces' weights were regularised as in E125, E129 and E132. E131, with weaker regularisation and chosen among many variants, came in 0.00005 worse than predicted, and we changed the rule after it. Final B (E136) landed 0.0000025 better than its prediction.

13 Reproducing it

Python 3.11 (`requirements.txt` pins every package), on Windows 11 or Linux, CPU only. Randomness is seeded: the simulator uses seed 20261020 for every draw, the misfit learner seed 20261031 (deterministic LightGBM), and the early files seed 20261011. The full sequence, with run times on our machine (Windows 11, 16 GB of memory), is in Appendix A. In short: download the package, build the per-tract features with one command (about 25 minutes), then `python src/final_build.py NAME` rebuilds a final file

from the package and the committed weights, compares it with the uploaded CSV, and checks that no excluded module was loaded. Both finals rebuild to within 3×10^{-16} (E134 2.0e-16, E136 3.1e-16).

13.1 What is not in the final files, and why

Early in the competition we built a few exploratory files that imposed another team's published table of mean scores at full strength, or that adjusted our file to match the public scorecard printed beside the leaderboard's top entry. We excluded those files and everything derived from them before building the finals: the finals are rebuilt from package-only pieces, and `final_build.py` stops with an error if the modules that did those things are loaded. The published averages we do use (the 16 cell means as calibration targets for the road scale, the 740 and 270 fire-station counts, the 0.052 places average, the fire-to-school ratio and the organisers' road-undefined counts) are listed with their sources in `src/published.py`. They set a handful of constants; none is a per-tract value.

13.2 Maintenance

If Overture releases change, re-run the feature scripts and `final_build.py`; the constants tied to published numbers are all in `src/published.py`. If the strata tables change, only the urbanity and tribal definitions in `src/strata.py` need checking. The simulator's fitted weights would need refitting only if the scoring formula changed.

14 Conclusion

As scored, most of the road gap is a property of the reference file's bookkeeping rather than of the map: a highway on a tract line is counted on both sides, and a stretch with several route numbers or names is counted several times. The effect is large, robust to how it is weighted, and uneven across places, so it does more than add noise: it bends the scorecard built to reveal inequity, understating how far rural and fire-exposed places fall behind and manufacturing most of the gap it reports for socially vulnerable ones. It can be removed on the reference side with the three changes of Section 6. Until it is, anyone who ranks places by the road score, or reads the scorecard's road-driven disparities, should first subtract the boundary and duplicate-record lengths, which the challenge package alone is enough to measure; the per-tract results we publish do this for every tract.

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A Reproducing every number

Every command runs from the repository root with `python` (Python 3.11).

Step	Command	Time
Quick check of Part A	`src/download.py --strata`, then `src/quick_check.py`	1 min
Download the package	`src/download.py`	40 min
Features per tract	`src/build_features.py` (every builder, in order)	25 min
Rebuild final B	`src/final_build.py e136_finalB` (rebuilds E131, E132, E134)	10 min
Rebuild final A	`src/final_build.py e134_learner`	8 min
Part A numbers	`src/writeup_numbers.py`, then with `displaced`, `scorecard`, `sensitivity`	5 min
Outside check, Section 4.5	`src/external_context.py` (add `fetch` to download again)	1 min
Figures	`src/paper_figures.py`	1 min
This PDF	`pdflatex paper.tex` (twice), in `docs/paper`	1 min

Expected outputs (the download time assumes about 3 MB/s):

- `docs/writeup/numbers.json`, with `road_gap_mean` = {as_scored 0.166, counted_once 0.047} and `added_ge_0.2_total.tracts` = 1548;
- `docs/writeup/worked_example.json`, with `route_highway_along_edge_m` = 1643.5.

B Data sources and licences

All data behind the scored files and Sections 2 to 4.4 come from the challenge package on Source Cooperative [2], downloaded on 10 October 2026 (the package version without the removed layers): Overture Maps release 2026-08-19.0 [5] (roads, buildings and rail under ODbL; places and infrastructure under CDLA-Permissive 2.0; © Overture Maps Foundation and its sources, including OpenStreetMap contributors), Census TIGER/Line 2025 tract geometry and ACS 2020–2024 housing units (public domain), and the strata tables listed in the package README under their own licences. No external data enters any scored file.

Outside the package (Section 4.5 only). Both sources are public and need no account or key. Adding `fetch` to `external_context.py` downloads them again; our extract for the 9,794 tracts is `external_tracts.csv` in `docs/writeup/`. The ACS tables [12] are U.S. Census Bureau data in the public domain. FEMA's index [13] is listed on data.gov as a U.S. Government Work and carries FEMA's terms, which ask for this statement: This product uses the Federal Emergency Management Agency's National Risk Index dataset API or downloadable datasets but is not endorsed by FEMA. The Federal Government or FEMA cannot vouch for the data or analyses derived from these data after the data have been retrieved from the Agency's website(s).

Section 4 also cites three government reports for context, each with its URL and retrieval date in the references [9, 10, 11].

C Glossary

Bookkept reference length the reference road length a tract receives under the organisers' overlay, including boundary highways and duplicate records.

Boundary highway a highway whose centreline is the tract line.

Duplicate record an extra reference record on the same stretch of road for a second route number or name.

Agreement with the hidden target the covariance between a change and the target, read from the score change and scaled like a correlation.

Road-scored tract a tract whose road component is defined.

Simulator our model that draws the hidden reference and applies the scoring formula.

Teacher a combination of scored files whose shape the package-only pieces are fitted to.

D Submissions and use of AI tools

Selected finals: `EVGrZBHt` (E134, public RMSE 0.05556589, private 0.056132715) and `E3u62GkV` (E136, public RMSE 0.055506023, private 0.05613082). We used an LLM coding assistant for code drafting, debugging and editing this document. Every analysis, number and claim was checked by the team, and the pipeline runs without it.